

A C^1 Tangent Cone for Density Matrices

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2026-07-18

Status

This packet gives a likely valid full solution to the regularity question raised in Cai–Govindarajan–Junge, arXiv:2504.04883. The source defines the tangent cone to the finite-dimensional density-matrix state space using one-sided C^2 curves, then says that a possible future direction is weakening the analytic condition to C^1 . The result below proves that this weakening is possible and that the cone is unchanged.

Even though the state space is convex, we see that the tangent cone at a corner point is not given by linear paths, we must consider all arbitrary C^2 paths if we want to understand the tangent space. this also shows that analytic considerations, whether we consider C^2 paths or C^n paths, may possibly affect the tangent cone at a corner point. A possible direction for future research is weakening the analytic condition to C^1 . It turns out that the C^2 condition is enough for the next characterization, which we will then connect to the time-irreversibility of quantum Markovian semigroup.

Theorem 3.3 ([CM20, HUL04]). Let $\rho \in \mathcal{D}(\mathcal{H})$ and $f : \mathcal{H} \rightarrow \text{supp } \rho$ be the projection onto the support of ρ , then

$$T_\rho^+ \mathcal{D}(\mathcal{H}) = \{x \in \mathbb{B}(H) : x = x^*, \text{tr } x = 0, (1 - f)x(1 - f) \geq 0\}$$

To prove that Theorem 3.3 works for all states in $\mathcal{D}(\mathcal{H})$ we first need the following diagonalization lemma.

Lemma 3.4 ([Pau02]). Let A be an invertible matrix. We write as a block matrix:

$$\begin{pmatrix} A & B \\ B^* & C \end{pmatrix} \geq 0 \iff A \geq 0, -B^* A^{-1} B + C \geq 0$$

Using this lemma, we can now show that our characterization holds for all states.

Proof of Theorem 3.3. Let $\rho_t \in C^2([0, \infty), \mathcal{D}(\mathcal{H}))$, for small t , we can write it as

Figure 1: Source page 14: future-work sentence and the existing C^2 tangent cone theorem.

Theorem

Let $\mathcal{H}$ be finite-dimensional and let

$$\mathcal{D}(\mathcal{H}) = \{\rho \in B(\mathcal{H}) : \rho \geq 0, \text{tr } \rho = 1\}.$$

For $\rho \in \mathcal{D}(\mathcal{H})$, define the one-sided C^1 tangent cone

$$T_\rho^{(1)}\mathcal{D}(\mathcal{H}) = \{\gamma'(0) : \gamma \in C^1([0, \varepsilon), \mathcal{D}(\mathcal{H})), \gamma(0) = \rho\}.$$

If s is the orthogonal projection onto $\text{supp } \rho$, and $q = 1 - s$, then

$$T_\rho^{(1)}\mathcal{D}(\mathcal{H}) = \{x \in B(\mathcal{H}) : x = x^*, \text{tr } x = 0, qxq \geq 0\}.$$

In particular, the C^1 tangent cone agrees with the C^2 tangent cone of arXiv:2504.04883, Theorem 3.3. Consequently the source's identity

$$T_\rho^+\mathcal{D}(\mathcal{H}) = \{L(\rho) : L \text{ is Lindbladian}\}$$

remains true when $T_\rho^+\mathcal{D}(\mathcal{H})$ is defined using C^1 curves.

Idea of the Proof

A C^1 path has the expansion $\gamma(t) = \rho + tx + o(t)$. On the kernel of ρ , positivity gives $q\gamma(t)q = tqxq + o(t) \geq 0$. Dividing by t and taking the limit gives $qxq \geq 0$. Conversely, every x satisfying this condition can be realized by a quadratic correction. The correction absorbs the off-diagonal block that prevents the linear path $\rho + tx$ from staying positive at boundary points.

Proof

First suppose $x \in T_\rho^{(1)}\mathcal{D}(\mathcal{H})$. Then there is a one-sided C^1 curve $\gamma : [0, \varepsilon) \rightarrow \mathcal{D}(\mathcal{H})$ with $\gamma(0) = \rho$ and $\gamma'(0) = x$. Since every $\gamma(t)$ is self-adjoint and has trace one, $x = x^*$ and $\text{tr } x = 0$. Also

$$\gamma(t) = \rho + tx + r(t), \quad \|r(t)\| = o(t).$$

Compress to $q\mathcal{H} = \ker \rho$. Since $q\rho q = 0$ and $\gamma(t) \geq 0$,

$$0 \leq q\gamma(t)q = tqxq + qr(t)q.$$

After division by $t > 0$ and passage to the limit, closedness of the positive semidefinite cone gives $qxq \geq 0$.

Conversely, assume $x = x^*$, $\text{tr } x = 0$, and $qxq \geq 0$. If $q = 0$, then ρ is positive definite and $\rho + tx \in \mathcal{D}(\mathcal{H})$ for all sufficiently small $t \geq 0$, so x is realized by a C^∞ path.

It remains to treat the boundary case. Write matrices relative to $s\mathcal{H} \oplus q\mathcal{H}$:

$$\rho = \begin{pmatrix} R & 0 \\ 0 & 0 \end{pmatrix}, \quad x = \begin{pmatrix} A & B \\ B^* & C \end{pmatrix},$$

where $R > 0$ on $s\mathcal{H}$, $C \geq 0$ on $q\mathcal{H}$, and $\text{tr } A + \text{tr } C = 0$. Choose $D \geq 0$ on $q\mathcal{H}$ so large that

$$D \geq 2B^*R^{-1}B.$$

For small $t \geq 0$, define

$$\rho_t = \begin{pmatrix} R + tA - t^2(\text{tr } D)R & tB \\ tB^* & tC + t^2D \end{pmatrix}.$$

This curve is polynomial in t , satisfies $\rho_0 = \rho$, and has derivative x at 0. Its trace is

$$1 + t(\text{tr } A + \text{tr } C) - t^2(\text{tr } D) \text{tr } R + t^2 \text{tr } D = 1,$$

because $\text{tr } R = \text{tr } \rho = 1$.

For all sufficiently small t , the upper-left block

$$R_t = R + tA - t^2(\text{tr } D)R$$

is positive definite and satisfies $R_t^{-1} \leq 2R^{-1}$ in the Loewner order. The Schur complement of R_t in ρ_t is therefore

$$tC + t^2D - t^2B^*R_t^{-1}B \geq tC + t^2(D - 2B^*R^{-1}B) \geq 0.$$

Thus $\rho_t \geq 0$ for all sufficiently small $t \geq 0$. We have constructed a C^2 path in $\mathcal{D}(\mathcal{H})$ with derivative x , hence certainly a C^1 path.

Combining the two inclusions proves the formula for $T_\rho^{(1)}\mathcal{D}(\mathcal{H})$. The final Lindbladian assertion follows by comparing this formula with Theorem 3.8 of the source, which identifies the same cone with $\{L(\rho) : L \text{ is Lindbladian}\}$.

Verification Notes

The only point where boundary singularity enters is the Schur-complement estimate. The t^2D term dominates the order- t^2 off-diagonal obstruction, while the $-t^2(\text{tr } D)R$ term preserves trace without destroying positivity of the support block. No compactness or infinite-dimensional closure issue is involved.

Novelty Check and Limitations

On 2026-07-18, the run indexes and web searches were checked for arXiv:2504.04883, “How Far do Lindbladians Go”, “C1 tangent cone”, “density matrices tangent cone support projection”, “positive semidefinite C1 tangent cone”, and “Lindbladian tangent cone”. The search found the source paper itself and general convex-analysis material, but no exact later answer to the source’s C^1 weakening direction.

This packet does not solve the broader lifting/retract problem for selecting Lindbladian generators continuously, smoothly, or analytically along arbitrary paths. It only proves that the tangent-cone definition can be weakened from C^2 to C^1 .

References

- [1] J. Cai, A. Govindarajan, and M. Junge, *How Far do Lindbladians Go?*, arXiv:2504.04883.